AfyaPlus - Cost-Optimised AI Deployment

Executive one-pager | Week 7 Capstone | Model: GPT-4o mini

Executive summary

Recommendation: proceed with API-hosted inference for the current AfyaPlus workload, with caching, batching, budget alerts, and periodic cost review. Reassess self-hosting only if sustained traffic and measured economics materially change.

Unit economics

Scenario	Requests / month	Cost / 1,000	Monthly cost
Baseline	10,000	$3.665	$36.65
10x traffic spike	100,000	$0.915	$91.50

The 10x scenario costs more overall, but the modeled unit cost falls because infrastructure and operations costs grow more slowly than request volume.

Two optimisation levers

Lever	Measured result	Business meaning
Response cache	50% hit rate; LLM-only cost / 1,000 fell from $0.018540 to $0.009270.	Repeated requests can avoid another model call, directly reducing token spend.
Batch API	10 HTTP requests -> 2; wall time 188.91 ms -> 50.91 ms (-73.05%).	Reduces HTTP overhead and improves throughput; token cost stayed unchanged.

10x spike plan

Scale the application layer, keep the cache enabled, route non-urgent work through batches, monitor token use and cache hit rate, and investigate spend when the monthly budget approaches the warning threshold.

Budget control	Risk	Mitigation
$50 monthly budget; warning at $40 (80%); alert at $50 (100%).	Unexpected traffic or low cache hit rate can increase spend.	Monitor request volume and token use, review cache hit rate, and investigate abnormal model calls before the ceiling is reached.

API-hosted vs self-hosted

Using the current model assumptions, API token cost crosses the estimated $450/month self-host cost at about **2.73 million requests/month**. This is a cost crossover only, not an automatic migration point. Reliability, staffing, GPU utilisation, security, and model quality still matter.

Go / no-go conditions

Go: continue with API-hosted inference while the service stays within the $50 monthly budget, meets latency and quality targets, and caching remains effective. **Reassess:** if monthly spend, sustained traffic, provider constraints, data-residency needs, or model quality requirements change materially.

Pricing basis: GPT-4o mini input $0.15 / 1M tokens; output $0.60 / 1M tokens. Baseline full cost includes modeled LLM, infrastructure, and operations. Cache figures are LLM-only estimates from the controlled stubbed fixture test. Local latency measurements are indicative, not guaranteed production performance.